

CE 517: Hydroinformatics

Course Code	CE 517
Course Name	Hydroinformatics
L-T-P-C	3-1-0-4
Prerequisites	None; however, familiarity with any computer programming language (e.g., Python, R, MATLAB, etc.) will be helpful during the course.
Intended for	B.Tech. (4th year), PG and Ph.D. students
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

1. Introduction: Introduction to Hydroinformatics, hydrology, water resources, Smart water management; Forecasting and Early warning system; Major sources of meteorological, climatic, and hydrologic data. (6 Hours)
2. Databases and Data Models Data life cycle, Data structures, Database management, Data Storage and retrieval, and use data from data models; CSV; NetCDF; Big Data; query aggregate and pivot data using Structured Query language (SQL), Entity Relationship Model; Introduction to Programming and computational tools Python, R, excel, etc. (8 Hours)
3. Data Analysis and Visualization Exploratory data analysis techniques; Introduction to data visualization tools; Basic and Specialized Visualization Tools; Visualization tools for geospatial data; interactive data visualization; Creating data dashboards. (8 Hours)
4. Geospatial Analysis Analysis of vector and raster datasets; Map Scale and projections; Introduction to GDAL; raster and vector conversions; Analysis and visualization of DEM; Watershed analysis and characterization; Watershed assessment and susceptibility/ vulnerability mapping; Raster querying; Stack Mosaic; Introduction to WebGIS, Introduction to Google Earth Engine and Microsoft Planetary Computer. (8 Hours)
5. Modelling and Simulations Time series analysis, Rainfall-runoff modeling; statistical analysis: Regression, probability distributions, interpolation, autocorrelation, hypothesis testing, frequency analysis, and return period estimation; Missing data; Hydrological modeling: Uncertainty and sensitivity analysis; Calibration and validation using Monte Carlo (MC), Markov Chain Monte-Carlo (MCMC), Maximum likelihood estimation (MLE), Shuffled Complex Evolution Algorithm (SCE-UA), Dynamically Dimensioned Search algorithm (DDS); Machine learning techniques in Hydrology. (12 Hours)

Text Books

1. Kumar, P., Folk, M., Markus, M., & Alameda, J. C., **Hydroinformatics: data integrative approaches in computation, analysis, and modeling**, CRC Press, 2005.
2. Remesan R., Mathew J., **Hydrological Data Driven Modelling: A Case Study Approach**, Springer, 2015.

References

1. Tomer, S. K., **Python in Hydrology**, Green Tea Press, 2011.
2. Beven, K. J., **Rainfall-runoff modelling: the primer**, John Wiley & Sons, 2011.
3. Lee, T., Singh, V. P., & Cho, K. H., **Deep Learning for Hydrometeorology and Environmental Science**, Springer, 2021.

Note: Some research papers, reports and handouts will also be provided as study material during the class.